Interdisciplinary project

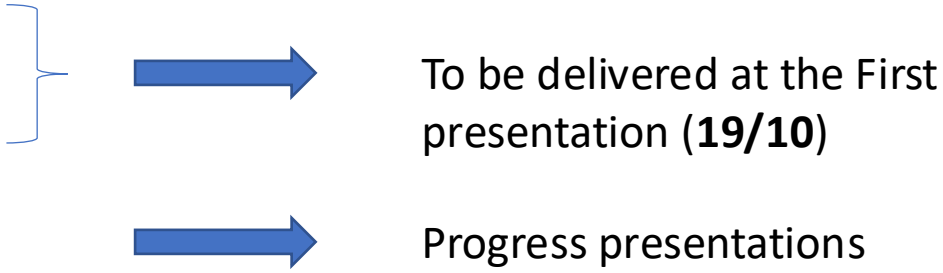


Project planning

ICT4SS

Project Management

- Complex project require organisation of the work
- The exercise done in this course is very similar to typical work projects to be developed by teams
- In this course you will have to organize your workflow according to the typical phases of collaborative projects
  - Workplan preparation
  - Milestones



How to organise the project

- In order to grant the success of a complex project, a good organisation is necessary to take into account
  - OBJECTIVES
  - EFFORT AND TIME REQUIRED
  - VERIFICATION POINTS
  - COMMUNICATION OF THE RESULTS

- Even if in a downscaled way, in this course we follow the typical project organisation of European projects

OBJECTIVES and REQUIREMENTS

- The main objective of the project must be **understood** and **split** in intermediate smaller objectives to achieve step by step
- In order to understand the objective you need to have clear **REQUIREMENTS**

USER REQUIREMENTS

FUNCTIONAL REQUIREMENTS

TECHNICAL REQUIREMENTS

- **Your project will be evaluated at the Final Review versus the requirements that you defined**
- **Requirement must be organized in these three different classes**

USER REQUIREMENTS

- The first project phase will consolidate the system objectives and challenges with the aim to detail the target performance and related operational environment, justification of the remaining operational limitations and constraints (if any).

- You must keep in mind what are the expectations of the users
- User Requirements might not be strictly technical or expressed in technical language

Who is the final user of your product?
What should a user do with your product?
How should he/she use it?
What would a user like to have as interface to your product?
…

FUNCTIONAL REQUIREMENTS

- The first project phase will consolidate the system objectives and target challenges with the aim to detail the functionalities

What should my product do overall?
Which detailed operations should be implemented to achieve the overall functioning?
How should it work?
In which conditions should it work?
…

TECHNICAL REQUIREMENTS

- The first project phase will consolidate the system objectives and target challenges with the aim to detail the technologies needed, the implementation choices

What software/hardware do we need?
What performance should we achieve?
What tools (software, simulation, labs) are needed to test the product?
…

Part of the requirements are already detailed in the topic description, but they are the starting point to be further detailed and understood

REVIEW OF THE STATE OF THE ART

- Definition of the requirements goes along with the review of what is already available in terms of technologies and solutions
- Analysis of the context in which your project is developed, the issues and solutions already implemented
- Similar products “on the market”
- Scientific literature

EFFORT AND TIME

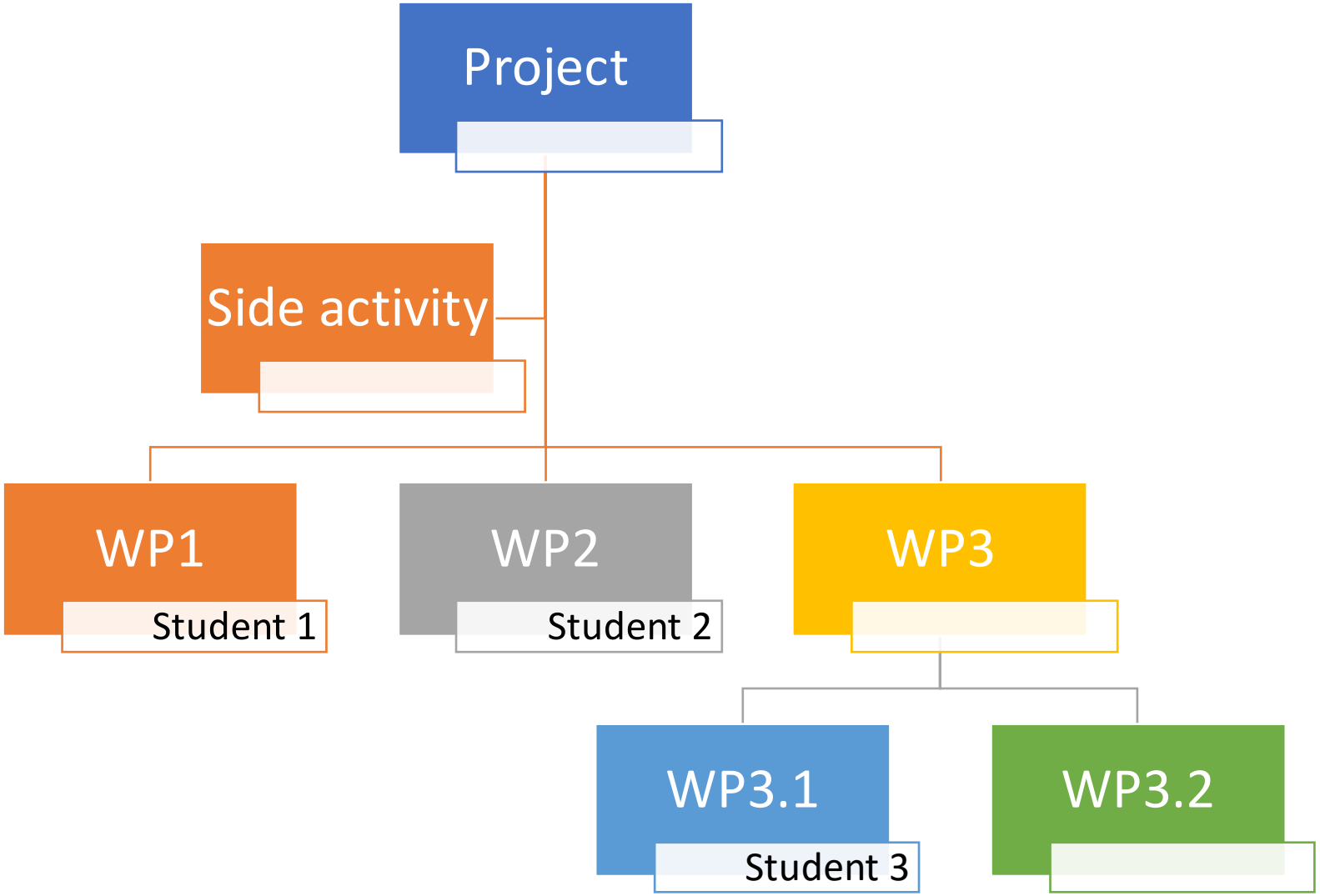
- Careful analysis of the problem is essential to plan the activities

- This can be done using a **splitting of the project in workpackages and tasks**, assessing the effort needed for each of the tasks
- The unit used to quantify the effort is typically the person/hours or person/month
  - Time-needed to perform the task
  - Hours spent by each person
    - *A task can be done in 2 weeks by 1 worker, but it may not be possible to do it in 1 week by 2 workers…*

  - Clearly define who is doing what
  - Balance the effort

  - Be realistic! (time available, exam sessions, etc..)

Work breakdown structure (WBS)



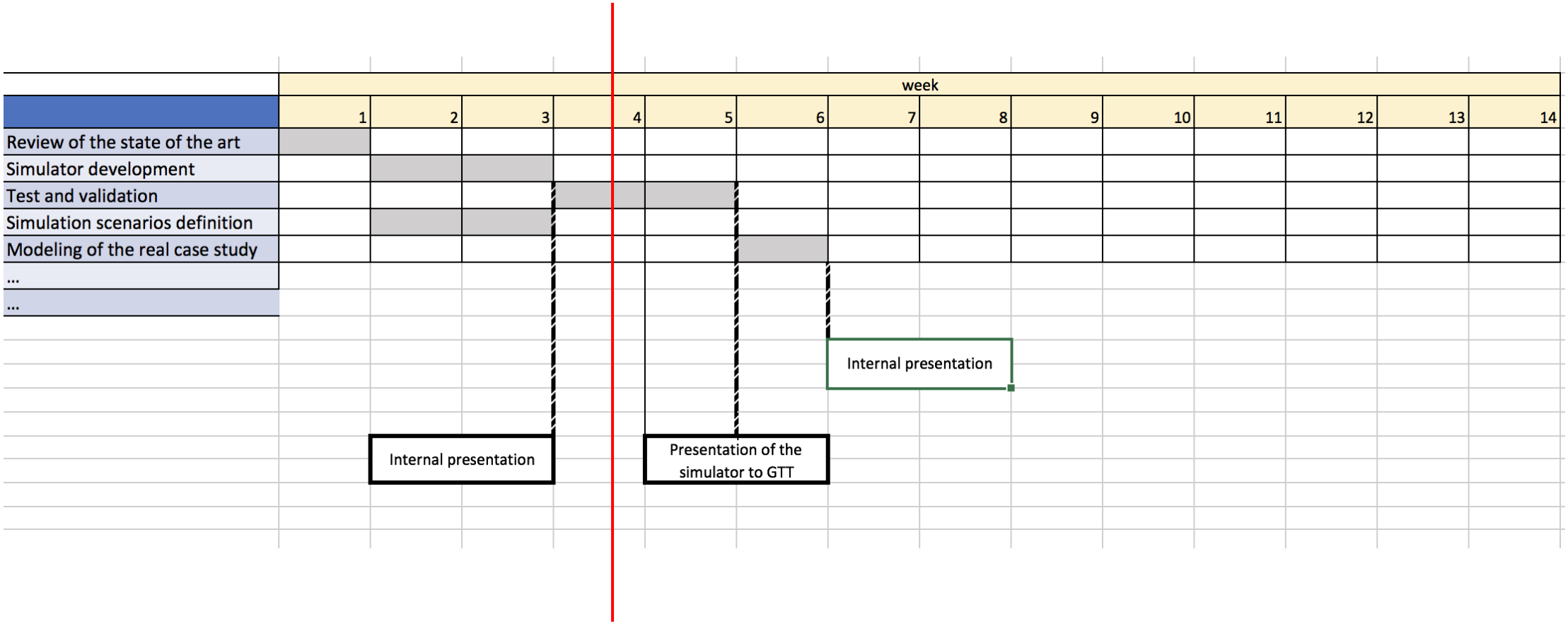
Workplan - schedule

- The WBS takes into account the slpitting of the effort but NOT the temporal distribution of it

- The temporal distribution have to be described in a different way

- Use of a Gantt chart is suggested, showing interdependencies between the workpackages and tasks
  - https://en.wikipedia.org/wiki/Gantt_chart

Example of Gannt chart



Gannt deviations

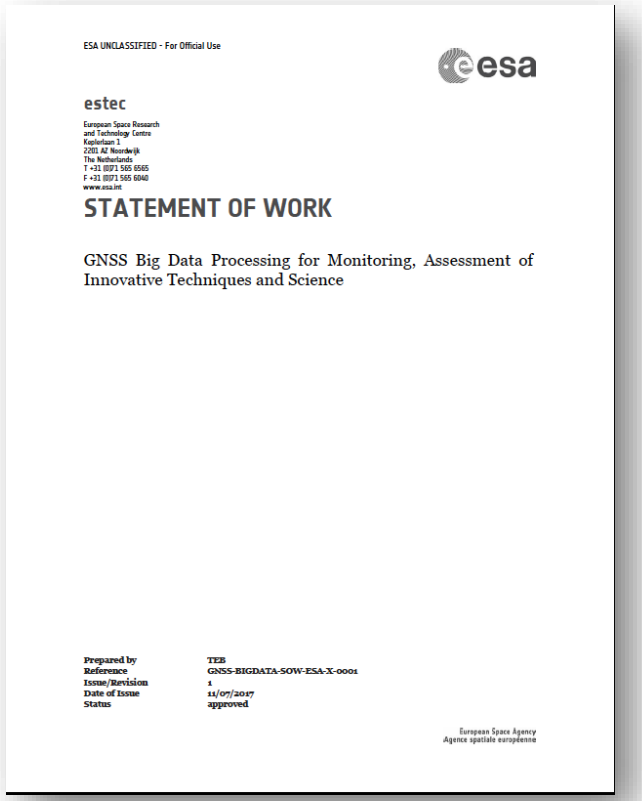
- This chart will be used as a reference for checking the advancement of the work and will be shown at each milestone to present the advancement of the work, comparing old and new versions in case of changes
- **Changing of the Gannt defined at the beginning of the project is a negative thing**! That's why the workplan should be carefully prepared

- Deviations from the plan can be accepted if they are **properly justified**
  - *«I didnt have time to do the work»* is not a justification
  - *«Considering the output of the PDR, perfroming the test we noticed that we had to change…»* may be acceptable

A real example

GNSS Big Data Processing for Monitoring Assessment of Innovative Techniques and Science

G-BIGDAT



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estec

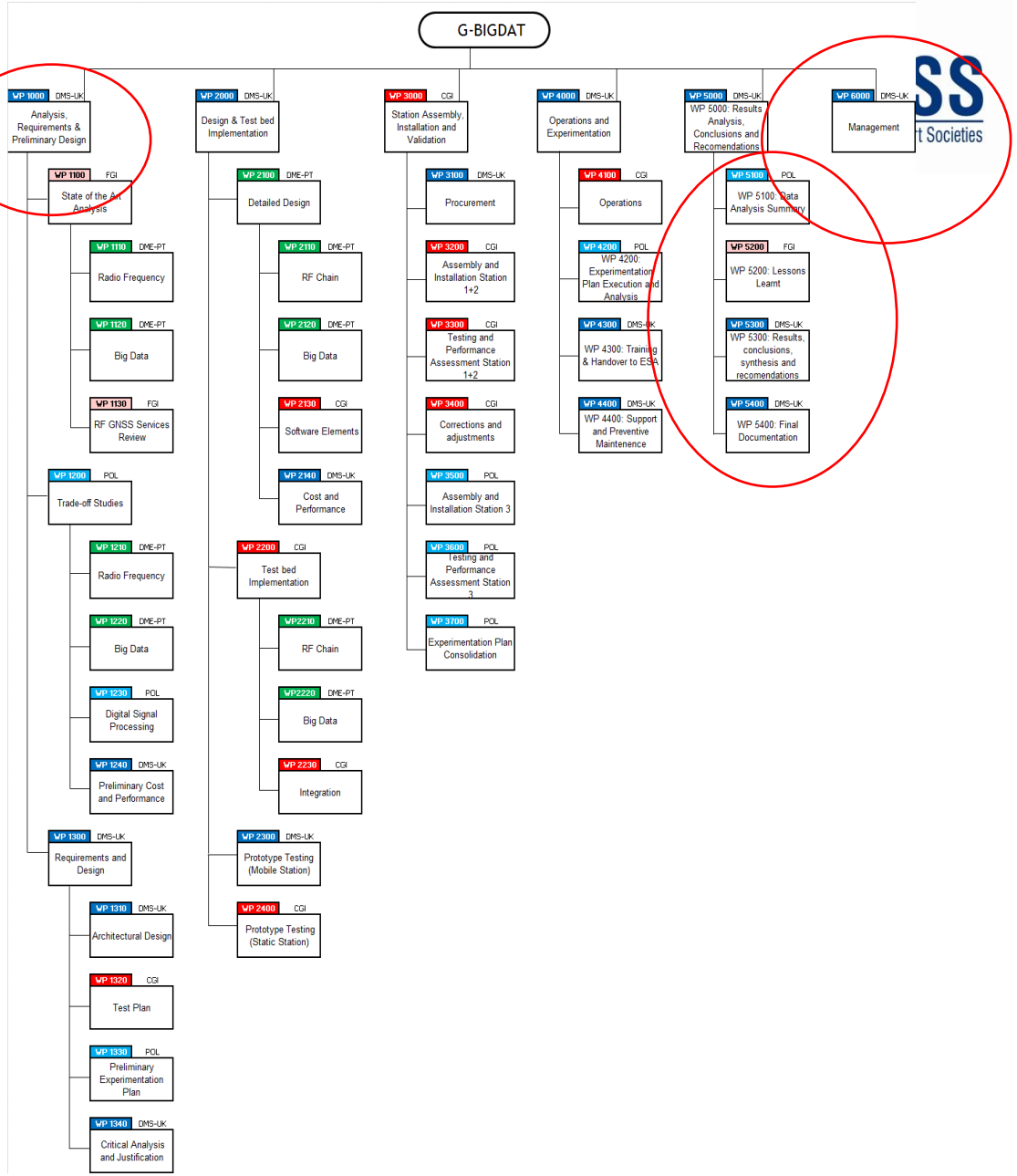
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STATEMENT OF WORK

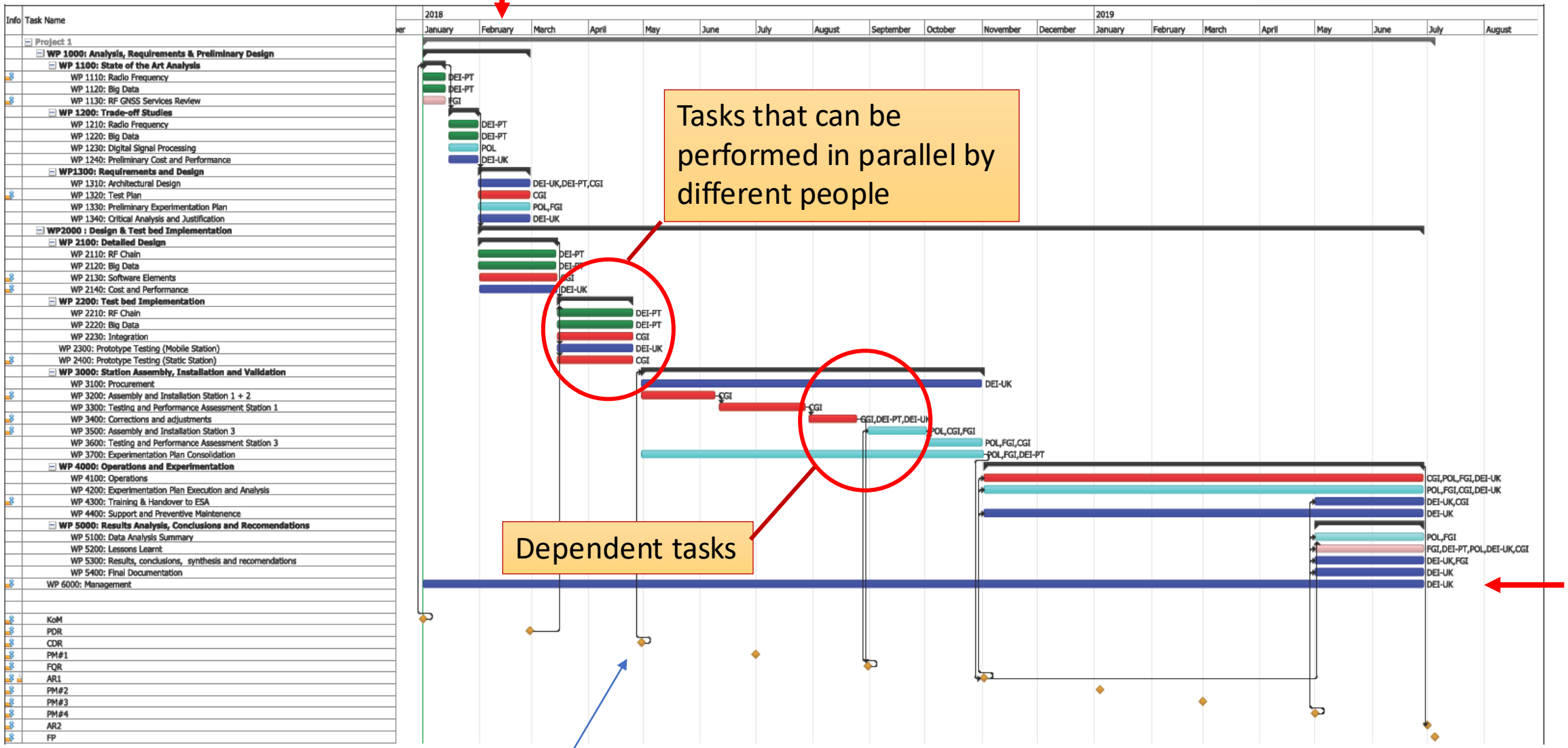
GNSS Big Data Processing for Monitoring, Assessment of Innovative Techniques and Science

Prepared by	TEB
Reference	GNSS-BIGDATA-SOW-ESA-X-0001
Issue/Revision	1
Date of Issue	11/07/2017
Status	approved

European Space Agency
Agence spatiale européenne



A real case


Info
Task Name
2018
January
February
March
April
May
June
July
August
September
October
November
December
2019
January
February
March
April
May
June
July
August
Project 1
WP 1000: Analysis, Requirements & Preliminary Design
WP 1100: State of the Art Analysis
WP 1110: Radio Frequency
WP 1120: Big Data
WP 1130: RF GNSS Services Review
WP 1200: Trade-off Studies
WP 1210: Radio Frequency
WP 1220: Big Data
WP 1230: Digital Signal Processing
WP 1240: Preliminary Cost and Performance
WP1300: Requirements and Design
WP 1310: Architectural Design
WP 1320: Test Plan
WP 1330: Preliminary Experimentation Plan
WP 1340: Critical Analysis and Justification
WP2000 : Design & Test bed Implementation
WP 2100: Detailed Design
WP 2110: RF Chain
WP 2120: Big Data
WP 2130: Software Elements
WP 2140: Cost and Performance
WP 2200: Test bed Implementation
WP 2210: RF Chain
WP 2220: Big Data
WP 2230: Integration
WP 2300: Prototype Testing (Mobile Station)
WP 2400: Prototype Testing (Static Station)
WP 3000: Station Assembly, Installation and Validation
WP 3100: Procurement
WP 3200: Assembly and Installation Station 1 + 2
WP 3300: Testing and Performance Assessment Station 1
WP 3400: Corrections and adjustments
WP 3500: Assembly and Installation Station 3
WP 3600: Testing and Performance Assessment Station 3
WP 3700: Experimentation Plan Consolidation
WP 4000: Operations and Experimentation
WP 4100: Operations
WP 4200: Experimentation Plan Execution and Analysis
WP 4300: Training & Handover to ESA
WP 4400: Support and Preventive Maintenence
WP 5000: Results Analysis, Conclusions and Recomendations
WP 5100: Data Analysis Summary
WP 5200: Lessons Learnt
WP 5300: Results, conclusions, synthesis and recomendations
WP 5400: Final Documentation
WP 6000: Management
KoM
PDR
CDR
PM#1
FQR
AR1
PM#2
PM#3
PM#4
AR2
FP
DEI-PT
DEI-PT
FGI
DEI-PT
DEI-PT
POL
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Tasks that can be performed in parallel by different people
Dependent tasks
meetings
milestones

An example from a previous Int. Project

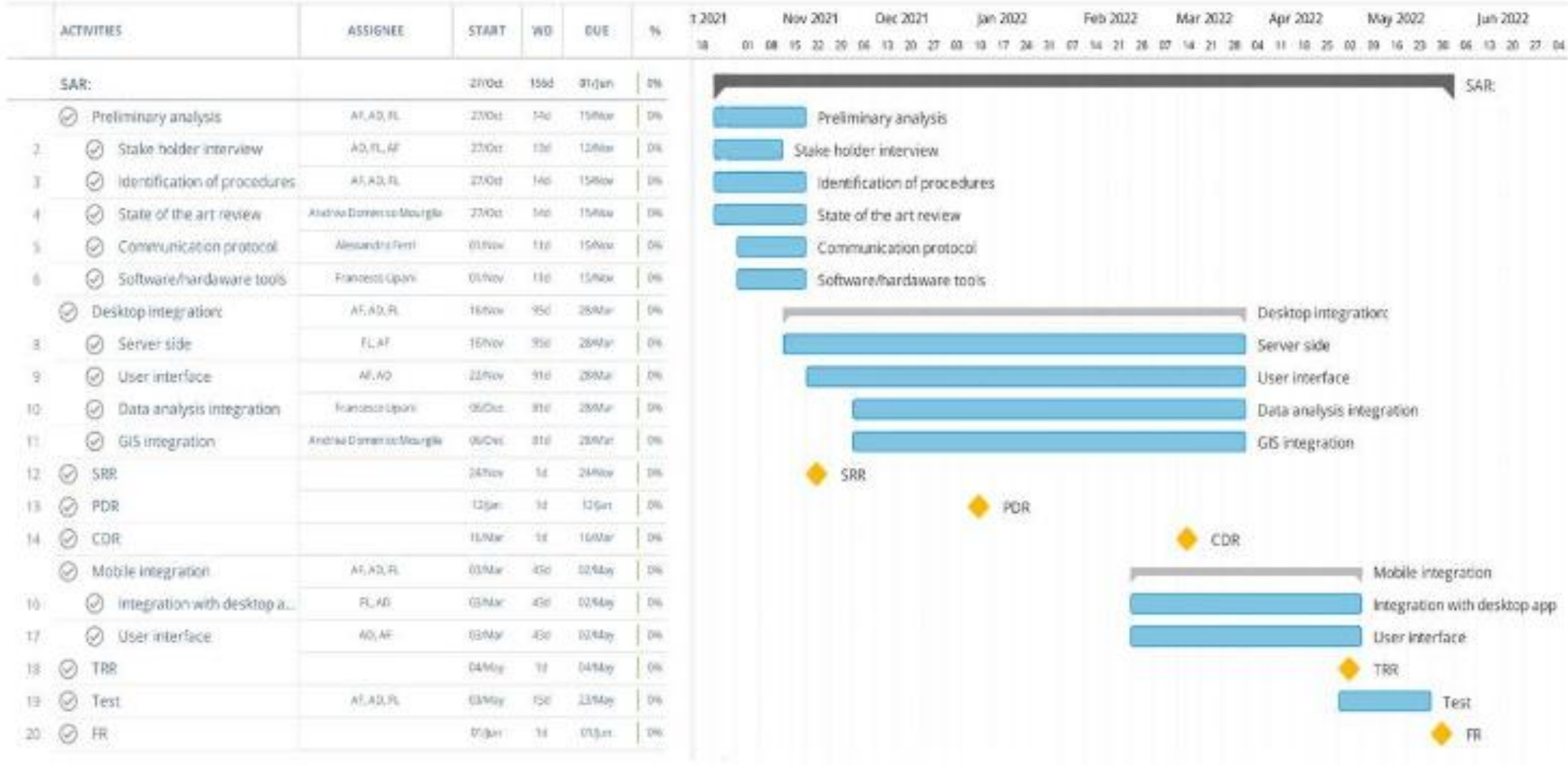


Figure 1: GANTT chart

Milestones

- Milestones are the verification point of the project development.
- They are usually associated to review meeting (in our case the presentation in the classroom)

- The project starts with a

  - *Kick-off meeting* → Presentation of the project objectives
    Presentation of the worksplan (WBS, Gannt)
    Preliminary understanding of the requirements

The milestones

MILESTONES

- *System Requirement Review* (SRR) → Fundamental to define your objectives and the organisation of the work. To be done before end of November. Delivery of the **System requirement document**
- *Preliminary Design Review* (PDR) → Presents a preliminary architecture of your systems, with maybe some open points and open implementation choices expected to be in the 1st semester for every team
- *Critical Design Review (CDR)* → Presents a **consolidated** architecture of your system, that is unlikely to be changed unless crticalities are found during the tests
- *Test Readiness Review* (TRR) → It can be placed before the final tests are performed, presenting the **test-plan. Delivery of the Test Plan**
- *Final Review* (FR) → General presentation of the project, Delivery of the **test results document** and possibly a demo

NOTE: Typical milestones of ICT and Space projects funded by the European Space Agency

SysReq document

- The system requirements document must be delivered at the SRR
- It will be used during the final review to assess the achivements
- it is useful to trace back each development choice to the requirements

	Project name	Revision	
		Date	
		Page	1 of 1

System requirement document

Abstract

Brief description of the project and the project goals

Requirements list

USER REQUIREMENTS	
UREQ_1	
UREQ_2	
UREQ_3	
....	

FUNCTIONAL REQUIREMENTS	
FREQ_1	
FREQ_2	
FREQ_3	
....	

TECHNICAL REQUIREMENTS	
TREQ_1	
TREQ_2	
TREQ_3	
....	

Test plan and results

- At the TRR you will have to present a planning of tests needed to verify the satisfaction of the requirements
  - Test title
  - Test description
  - Which requirements are being verified
  - KPI e success/fail thresholds

- This table will be completed with the results of such test

- *A template table will be provided*

COMMUNICATION

- **Dissemination activities,**
  - **Presentations of the progress of the work**
  - **Final presentation**

- **Commercialisation activities/ Exploitation Plan**: set of actions taken by to undertake to prepare the ground for commercialisation of the results, in accordance with the approved Business plan, or further develop your project

The preparation of a video or a brochure for the FR is appreciated

Presentations

- Each milestone is not just a presentation of the work done (or not done) in chronological order
- Each Milestone has it’s own characteristics and objectives
- **ALWAYS refer to the defined requirements (USER, FUNCTIONAL and TECHNICAL)**

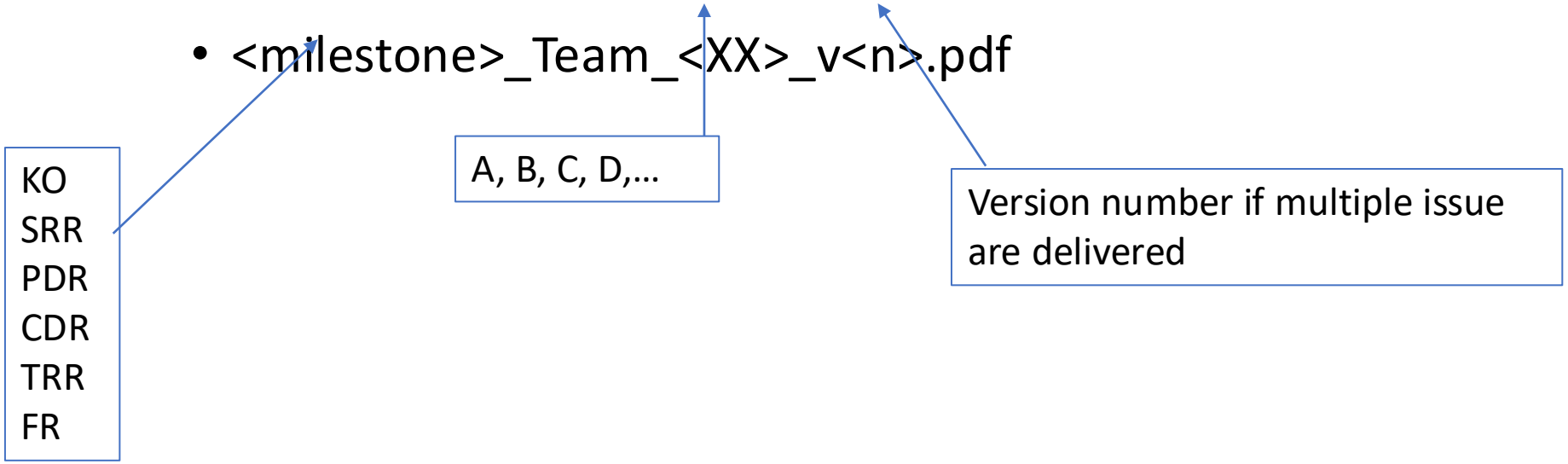
- Each presentation MUST contain:
  - The Gannt chart, highlighting variations if any
  - The list of meetings with your tutors

Each presentation MUST be uploaded **in pdf format** on the portal

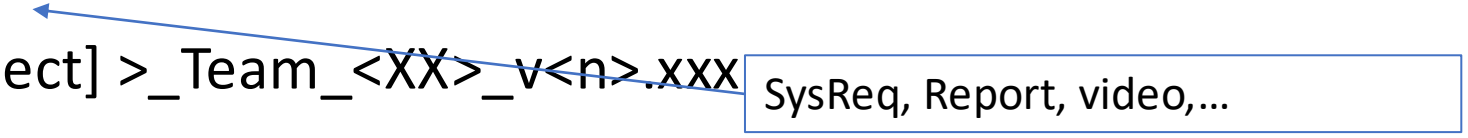
NOTHING HAS TO BE SENT VIA EMAIL TO THE TEACHER

Upload and file naming

- Files uploaded have to be named according to the following convention
  - <milestone>_Team_<XX>_v<n>.pdf



- Other files
  - [object] >_Team_<XX>_v<n>.xxx



WORKPLAN PREPARATION for the Kick-off

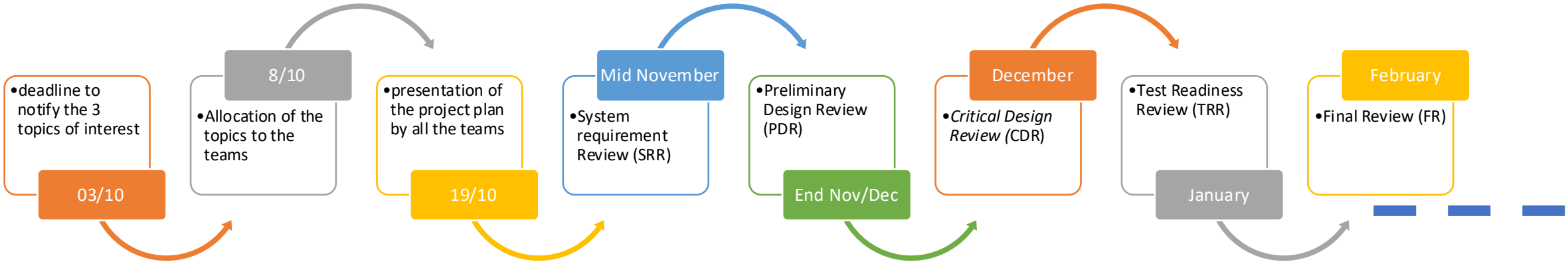
ICT4SS
Information and Communications Technologies for Smart Societies

On October 19, each project will have the Kick-Off.

In preparation to you have to prepare the first presentation including:

- Your understanding of the project topic and a review of the state of the art
- a **detailed workplan** that will guide you throughout the project development
- **a first set of requirements** that will be consolidated at the SRR
- a **project WBS** with allocation of the work to each member of the team
- **a Gaant chart** including
  - The time allocation of the WPs (start, end duration)
  - the milestones (SRR, PDR, CDR, TRR,FR), i.e. your progress presentations

Schedule of progress presentations



This is a <u>possible</u> schedule for a project to be developed in the first semester

Only the **19/10 date is fixed**, you have to decide when to put the other milestones according to your workplan

Classes will be allocated according to the presentations (corresponding to the milestones)

RULES FOR THE PRESENTATION

max 7 minutes per group

All the members of the group MUST participate to the presentation

Time for questions after the presentation by teachers and classmates

If someone is not present there must be a clear justification and must be notified in advance to prof. Dovis!

Nice quality of the template is appreciated, but the **content** is important
Be effective in your communication: focus on the **main messages** you want the audience to get

A possible outline of the first presentation

- Project's objective and goals
- State-of-the-art / background
- Workplan
- Preliminary requirements (user, functional, technical)

State-of-the art / background and requirements

- Give to the audience the essential background elements to understand you work

- Define the specific **requirements** of your project (preliminary version)
  - USER REQUIREMENTS : The project has to…
    - …
    - …
    - …
  - FUNCTIONAL REQUIREMENTS:
  - TECHNICAL REQUIREMTS: technical choices
- Requirements will be consolidated at the SRR

Project management requires discipline

- **The project management is not a nice touch “on top” of the work, but the organisation of the work itself, to get it done in a more effective and efficient way**

- A clean and smooth management of the project requires to manage the tools, be precise in the documentation of the work, follow the rules
- For these reasons, penalty will be applied to the final grade for
  - wrong naming of the files (-0.2 pts)
  - File sent via email to the teacher (-0.2 pts)
  - Missing member of the team at the milestone presentation without proper justification (-0.5 pts)
  - Presentations longer than 10 minutes (depending on the exceeded time)

Warm suggestions

- You are expected to attend the presentations of your classmates, this is useful to learn best-practices and have good ideas
- Do not focus too much on the fancy presentation templates. Focus on the content
- Reharsal the presentations, and share them with your tutors in advance

- Have fun with your projects!



Questions ???

Prof. Fabio Dovis

fabio.dovis@polito.it